

- 5 (a) State and explain one advantage of using alternating current when transmitting electrical energy.

.....

.....

.....

.....

[2]

- (b) An a.c. power supply is connected to a resistor R , a diode and a battery of e.m.f. 12.0 V as shown in Fig. 5.1.

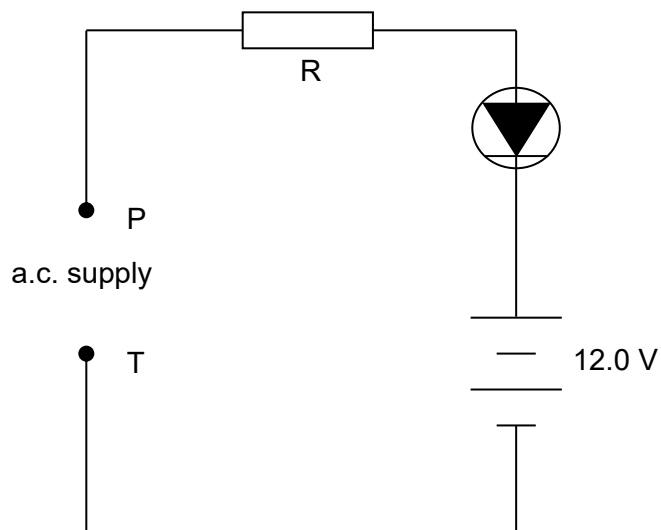


Fig. 5.1

The variation with time of the potential of P with respect to T, V_{PT} , is as shown in Fig. 5.2.

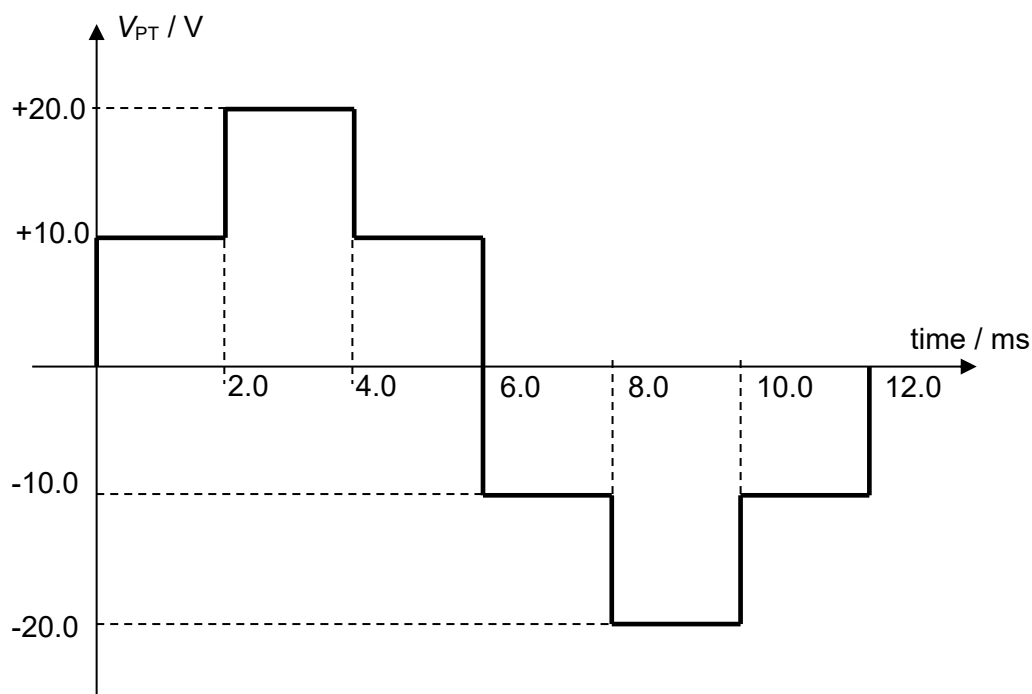


Fig. 5.2

- (i) Calculate the root-mean-square potential difference V_{rms} across points P and T.

$$V_{\text{rms}} = \dots\dots\dots \text{ V} \quad [2]$$

- (ii) Determine the potential difference across resistor R when

1. $V_{\text{PT}} = +10.0 \text{ V}$,

$$\text{potential difference} = \dots\dots\dots \text{ V} \quad [1]$$

2. $V_{\text{PT}} = +20.0 \text{ V}$.

$$\text{potential difference} = \dots\dots\dots \text{ V} \quad [2]$$

- (iii) Hence, sketch in Fig. 5.3 the variation with time of the potential difference across resistor R, V_{R} .

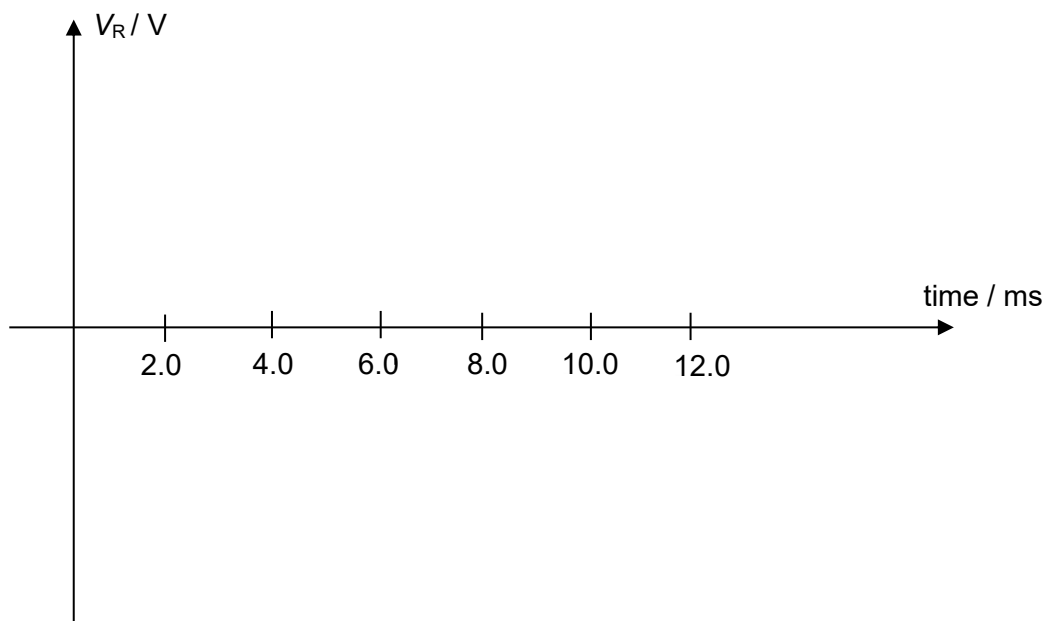


Fig. 5.3

[2]

[Total: 9]